



USB 20W COB LED Table Lamp

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Most table lamps available in the market do not provide sufficient light to read small-size letters and fonts, especially those found on electronic components. This 20-watt chip-on board (COB) LED table lamp gives sufficient light of about 270 lumens brightness.

Each COB actually has several LED (light emitting diode) chips mounted together on the same substrate, which is often ceramic or metal.

The lamp can be plugged in to any 5V USB adaptor or mobile charger, has light dimmer control, and can replace the circuit in an existing LED fixture. The proposed enclosure for the USB 20W COB LED table lamp is shown in Fig. 1.



Fig. 1: Proposed table lamp for the project

Circuit and description

Circuit diagram of USB 20W COB LED table lamp is shown in Fig. 2. The circuit includes a COB LED (LED1), 555 timer IC1, TIP32C power transistor T1, signal diodes D1 and D2, and XL6009 DC-DC converter/booster module.

COB LEDs (Fig. 3) are power-

ful yet quite handy for making LED lights. Most COB LEDs come with heat-sink and are ready to use. You just need a suitable driver and a pair of wires.

COB manufacturers release data sheets for their products that dictate each unit's specifications as well as minimum, typical, and maximum values for a number of different variables. A few examples of a COB's characteristics that can be found in the data sheet include colour temperature, typical forward voltage, maximum current, temperature, luminous flux, and relative spectral distribution.

There are different types of COB LEDs, and any suitable type can be used in the circuit. You may refer to 'Choosing The Right COB LED' article on <https://www.electronicsforu.com/buyers-guides/hardware-buyers-guide/led-cob-buyers-guide>

555 timer IC is configured in an astable mode with pulse-width modulation (PWM) control. Potmeter VR1 (470-kilo-ohm) is used to control the brightness of the COB LED through diodes D1 and D2.

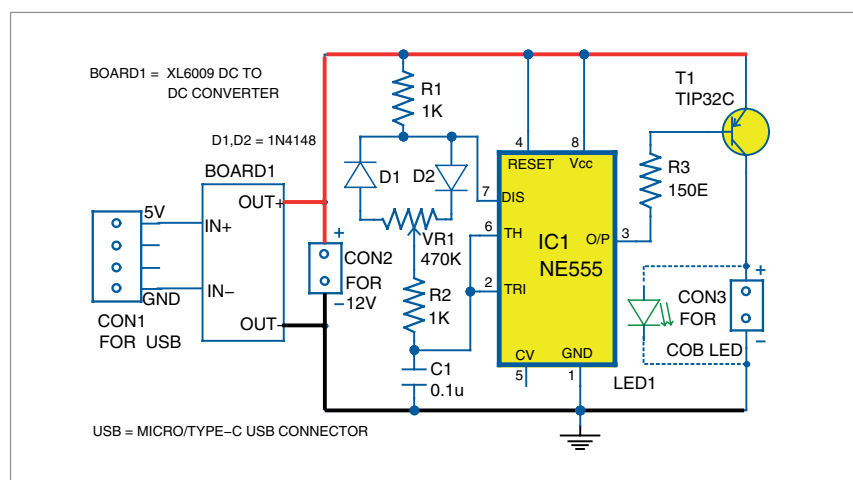


Fig. 2: Circuit diagram of COB LED table lamp



Fig. 3: COB LEDs